

Here you are using open pdk osu35.

<https://www.vlsitechnology.org/html/libraries04.html>

DFF in standard cell library has reset. So you should not neglect reset input in module definition. Copy the .lib file and .v file of pdk to your working folder fault_environment. These files are uploaded in lms

RTL Design

```
module counter(input clk, input rst, output reg [3:0] Q);
```

```
    always @(posedge clk) begin
```

```
        if (rst)
```

```
            Q<=0;
```

```
        else
```

```
            Q <= Q + 1;
```

```
    end
```

```
endmodule
```

Synthesize it -Yosys script

```
read_verilog counter.v
```

```
read_liberty -lib osu035_stdcells.lib
```

```
synth -top counter
```

```
dfflibmap -liberty osu035_stdcells.lib
```

```
proc; opt; flatten; opt
```

```
techmap
```

```
opt
```

```
stat
```

```
write_verilog counter_trial.v
```

After Synthesis **Add scan chain** (To see the options type **fault chain --help**)

```
fault chain counter_trial.v --clock clk --reset rst --liberty osu035_stdcells.lib -o  
counter_scan1.v --sout sout
```

Cut scan-chain for ATPG Prepares the netlist for ATPG simulation remove feed backs

(To get the options fault cut --help)

```
fault cut counter_scan1.v --clock clk -o counter_scan_cut.v
```

Run ATPG to generate test vectors- (To get the options `fault atpg--help`)

```
fault atpg counter_scan_cut.v --cell-model osu035_stdcells.v --clock clk -o patterns.tv.json  
--output-coverage-metadata coverage.yml
```